Question no.1
Find the equation of the line that passes through the point *(6, 7)* and is perpendicular to line *2x + 3y = 8.*
Solution:
The line *l* can be written as *y = -2x/3 + 8/3.*
So its slope is -2/3.
Slope of the line perpendicular to *l* is 3/2.
The equation is *y - 7 = 3/2(x - 6)*

Question no.2

Find the slope of the line through the points (-5, -2) and (-6, 7)

$$\text{slope} = m = \frac{rise}{run} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-(-2)}{-6+5} = \frac{9}{-1}$$

Question no.3
Find slope of the line $6x - 8y = 26.$

$$6x - 8y = 26$$

$$-8y = -6x + 26\,.$$

$$y = \frac{6}{8}x - 26/8$$

Slope is 6/8 or3/4
Question no.4

Find the point of intersection of the lines $2x + 3y = 7$ and $3x + 5y = 3$.

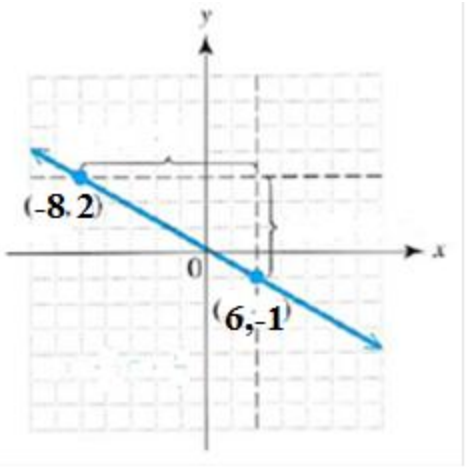
Solution
To find the point of intersection we need *(x, y)* which lie on both the lines, i.e. which satisfy the two equations simultaneously. We need to solve the equations simultaneously. Solving the equations simultaneously we get *x = 26*and *y = -15.*

Question no.5

Find the equation of the perpendicular bisector of the line joining the points A(2, 8) and B(6, 2).
Let *M* be the mid-point of *AB.* Then M(4,5)
Gradient of *AB* is *(8 - 2)/(2-6) = 6/-4=-3/2* .

So slope of required line is 2/3 . Equation of required line is *y - 5 = 2/3(x - 4)*
Question no.6

Find the slope for the following graph:



$$\text{slope} = m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$
$$= \frac{-1-2}{(6)-(-8)} = \frac{-3}{14}$$

Question no.7

Find the center and radius of the given equation of circle:
$(x+5)^2 + (y-3)^2 = 36.$

$\because$ The equation of a circle with center (h, k) and radius r in standard form
$(x-h)^2 + (y-k)^2 = r^2$
$\Rightarrow$ Comparing the given equation with standard form, we see that
$(x-(-5))^2 + (y-(3))^2 = 6^2$
$\therefore$ The center is $(-5, 3)$ and radius is 6.

Question no.8
Are the lines $4x+5y=6$ and $5x-4y=2$ parallel, perpendicular, or neither?

$\because$ $y = mx + c$; where $c = y_1 - mx_1$

For Line:1 $4x + 5y = 6$

$\Rightarrow$ $5y = -4x + 6$

$\Rightarrow$ $y = -\frac{4}{5}x + \frac{6}{5}$

$\Rightarrow$ slope $= m_1 = -\frac{4}{5}$

For Line:2 $5x - 4y = 2$

$\Rightarrow$ $-4y = -5x + 2$

$\Rightarrow$ $y = \frac{5}{4}x - \frac{2}{4}$

$\Rightarrow$ slope $= m_2 = \frac{5}{4}$

$\because$ Two lines with gradients (slopes) m_1 and m_2 are perpendicular if $m_1.m_2$ = -1, or $m_1 = -\frac{1}{m_2}$.

$\therefore m_1.m_2 = \left(-\frac{4}{5}\right)\left(\frac{5}{4}\right) = -1$

Question no.9

Given the equations of the lines $3x + y = -4$ and $3x + 3y = 1$. Do these lines intersect? If they do, find the point of intersection.

$\because$ $y = mx + c$; where $c = y_1 - mx_1$

For Line:1 $3x + y = -4$

$\Rightarrow$ $y = -3x - 4$

$\Rightarrow$ $y = -3x - 4$

$\Rightarrow$ slope $= m_1 = -3$

For Line:2 $3x + 3y = 1$

$\Rightarrow$ $3y = -3x + 1$

$\Rightarrow$ $y = -x + \frac{1}{3}$

$\Rightarrow$ slope $= m_2 = -1$

Since these two lines have different gradients (slopes), they must intersect.

To find the point of intersection we need (x, y) which lie on both the lines,

i.e. which satisfy the two equations simultaneously.

We need to solve the equations simultaneously.

Solving the equations simultaneously we get x = -13/6 and y = 5/2

So the point of intersection is $(-13/6, 5/2)$.